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Paper A Recall June 2016 - EMI Questions

by SPMM Administrator - Thursday, 24 November 2016, 1:27 PM

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Extended Matching Questions

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EMI Theme: Personality disorders

Options

A.Narcissistic personality disorder

B.Borderline personality disorder

C.Schizoid personality disorder

D.Schizotypal personality disorder

E.Dissocial personality disorder

F.Dependent personality disorder

G.Histrionic personality disorder

Lead In: What is the most likely diagnosis for each of the following patients?

1)Todd is a 25-year-old male who works in a family-run corner shop. He forgot to lock-up after work, which resulted in the store being burgled and a financial loss for the business. Todd appeared to demonstrate no concern for his actions. This is not the first time he has been careless and he is generally known to be unreliable.

2)Sabine is a 19-year-old fashion student. She holds the self-perception of being unique and superior to her peers. She told her supervisor that she felt she was exceptional and that was going to be the 'next Chanel', despite scoring a satisfactory grading in her first year.

3)Seema is a 33-year-old doctor who irritates her friends by speaking over them at dinner parties. Her friends complain that she 'always has to be the centre of attention.'

4)Jason is a 45-year-old courier. He has a poor rapport with his work colleagues and tends to withdraw socially. Despite working for his company for fifteen years, he has never attended an out-of-work social gathering.

5)21-year-old history student, Raphael, met up with his dissertation supervisor, who was baffled to learn that Raphael did not appear to be remotely bothered with the feedback he received.

Answers: 1-E, 2-A, 3-G, 4-D, 5-C

Explanation: 1) In dissocial personality disorder, individuals show a lack of concern for others, irresponsibility, irritability, aggression, inability to maintain enduring relationships and disregard and violation of others' rights 2) In narcissistic personality disorder, grandiosity, lack of empathy and the need for admiration are common traits 3) In histrionic personality disorder, individuals show a high-degree of self-dramatization, shallow affect, egocentricity, craving attention, excitement and manipulative behaviour 4) Interpersonal discomfort with peculiar ideas, perceptions, appearance and behaviour are common traits in schizotypal personality disorder 5) Emotionally cold, detachment, lack of interest in others, indifference to praise or criticism, excessive introspection and fantasy are common traits of schizoid personality disorder.

(Ref. Oxford Handbook of Psychiatry)

EMI Theme: Switching antidepressants

Options

A.Cross taper with caution

B.Stop and start other drug immediately

C.Withdraw and wait for 1 week

D.Withdraw and wait for 2 weeks

E.Withdraw and wait for 5-6 weeks

F.Half dose, start new drug, then slow withdrawal

G.Concomitant prescription is permissible

Lead In: What is the advised route for the following changes in medication?

1)Trazodone to Mirtazapine

2)Fluoxetine to Tranylcypromine

3)Phenelzine to Sertraline

4)Amitriptyline to citalopram

5)Fluoxetine to duloxetine

Answers: 1-A, 2-E, 3-D, 4-F, 5-B

When swapping from one antidepressant to another, abrupt withdrawal should usually be avoided. Cross tapering is preferred, where the dose of the ineffective or poorly tolerated drug is slowly reduced while the new drug is slowly introduced. 1) Cross taper with caution when you wanted to switch from Trazodone to Mirtazapine. 2) When you switch from Fluoxetine to Tranylcypromine, withdraw and wait at least 5 weeks, as Fluoxetine has the longest half-life. 3) When you switch from MAOI like Phenelzine to SSRI like Sertraline, withdraw and wait for 2 weeks. MAOIs should not be co-administered with SSRIs, as this combination could precipitate serotonin syndrome. 4) When you wanted to change from Tricyclics like Amitriptyline to Citalopram, first halve the dose of Amitriptyline, then add citalopram and then slowly withdraw Amitriptyline. 5) Abrupt switch is possible from fluoxetine to Duloxetine. (Ref: The Maudsley Prescribing Guidelines - 11th edition; Pg 271)

EMI Theme- Sleep disturbance

Options

A.Non-organic sleep disturbance

B.Night terrors

C.REM sleep disorder

D.Cataplexy

E.Akathisia

F.Hypersomnolence

G.Nightmares

Lead In: Which of the following best describes the following situation?

1)Susie is a 27-year-old nurse. She has a three-month-old son who cries during the nighttime and, as a result, has poor sleep. She complains of feeling excessively tired during the day.

2)George is a 53-year-old male who was recently prescribed Aripiprazole for his schizophrenia. His wife complains that he can't keep his legs still at night and that he tends to kick her in bed

3)Shilpa, an 8-year-old girl, wakes up repeatedly in the night, fearful and feeling that she is going to die

4)Juliana is a 46-year-old lady who suffers from acute episodes of collapse on laughter.

5)You meet Vanessa, a four-year-old girl, in your CAMHS outpatient clinic. Her mother tells you that Vanessa regularly wakes up distressed, complaining that she has had scary dreams.

Answers: 1-F, 2-E, 3-B, 4-D, 5-G

1) Hypersomnolence: recurrent episodes of excessive daytime sleepiness. It can be caused by chronic sleep deprivation, whereby an individual fails to get sufficient sleep for at least three months. This case is an example of secondary hypersomnia. 2) Akathisia is a movement disorder characterized by restlessness and an urge to be in constant motion. Behaviourally, this manifests as rocking while sitting or standing, marching on the spot, and the crossing and uncrossing of legs whilst sitting down 3) During night terrors, there are high levels of autonomic arousal several times at night. It is a sleep disorder in which the individual experiences feelings of terror or dread. 4) This is a case of cataplexy, a condition whereby strong emotions or laughter cause a person to suffer sudden physical collapse. The patient remains conscious. Low orexin levels can be one of the common risk factors. 5) A nightmare is an unpleasant, 'bad dream' that causes a strong emotional response.

EMI Theme: Adolescent Identity Development.

Options

A.Identity Moratorium

B.Identity Foreclosure

C.Identity Theft

D.Identity Achievement

E.Identity Diffusion

F.Identity Confusion

Lead In: Match the stages of adolescent identity development based on Marcia's structural interview

1.An 18-year-old man is in the process of developing a sense of identity when he leaves home to join the military. Here he has the opportunity to learn and develop new skills, which later on in life are incorporated into his sense of self.

2.A 17-year-old woman spends most of her time outdoors, in the company of a mixed group of young people. They spend time drinking alcohol, smoking cannabis; they have no plans for the future.

3.A 17-year-old man is given the choice by his parents to stop attending church if he wishes to. He doesn't wish to make an exploration of his own, but continues to attend church with his family, and to conform to their way of practicing religion.

4.A 21-year-old man after a lengthy period of exploration and self-searching, feels that he has a steady and stable sense of self. He finds himself committed to his work, with a clear sense of his goals and beliefs.

Answers: 1-A, 2-E, 3-B, 4-D

1) Identity Moratorium: This is a stage in which identity achievement is postponed whilst an intense exploration, which may be experienced as a crisis. At this point the young person is not yet ready to make a firm commitment. 2) Identity Diffusion: Identity issues have not been considered in detail and no firm commitments have been made for the future. During this stage the individual does not commit to any goals; it is characterized by procrastination, avoidance of responsibility and is the opposite of identity achievement.3) Identity Foreclosure: Identity issues have not been considered seriously but future commitments have been made in spite of this. In this stage teenagers accept and adopt the traditional values and cultural norms of their parents or surrounding culture, rather than exploring and forming their own.4) Identity Achievement: Various alternatives have been carefully considered and firm future commitments have been made. In this stage the individual is said to have solved their identity crisis and is able to make commitments to goals, values and beliefs.
(Ref:https://moodle.gllm.ac.uk/pluginfile.php/94785/mod_resource/content/1/Amanda_Tillson/Unit_4/Book_chapter_A

EMI Theme- ECG

Options

A.Tall tented T waves

B.Widened QRS Complexes

C.Reduction in QTc interval

D.Saddle-shaped ST elevation

E.Torsades de Pointes with use of antidepressant

F.QTc prolongation of >510ms

G.T wave inversion

Lead In: Match the following ECG changes with the most likely scenario;

1.A severely depressed man is prescribed a medication, which leaves him with a dry mouth. A few days later he is found at home collapsed, having taken a significant overdose.

2.A 50-year-old woman has been feeling low in mood, and finding it difficult to sleep. She is prescribed a medication by her GP, which is increased over a period of 4 weeks. Subsequently she has an ECG after reporting palpitations and dizziness.

3.A young man with first episode psychosis is prescribed a medication and three months later has a routine ECG.

4.A middle-aged man admitted to the ward with chronic psychotic illness reports some chest discomfort. An ECG is performed.

Answers: 1-F, 2-E, 3-C, 4-D

1) Amitriptyline is an antidepressant medication that causes dry mouth and QTc prolongation leading to severe ventricular arrhythmias, collapse and death. 2) Mirtazapine is an antidepressant medication. On a higher dosage, ECG may show Torsades de Pointes with patients reporting symptoms of dizziness and palpitations 3) Aripiprazole is an antipsychotic medication used for the treatment of first episode psychosis and it has been associated with a reduction in QTc interval 4) Clozapine is used for the treatment of chronic psychosis and it can cause myocarditis. Saddle-shaped ST elevation is seen in ECG.

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Paper A Recall - Part 3

by SPMM Administrator - Wednesday, 23 November 2016, 9:27 PM

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76. 'Driving a car' is an example of what type of memory?

A.Declarative

B.Semantic

C.Episodic

D.Working

E.Long term memory

Answer: E

Driving a car is an example of procedural memory, which is classified under non-declarative memory. Non-declarative memory is a component of long-term memory.

77. According to Kohlberg, which one among the following represent Conventional Stage?

A.Social Contract

B.Authority

C.Universal Ethics

D.Obedience

E.Punishment

Answer: B

Social order or Authority Orientation represents conventional stage in Kohlberg's theory of moral development. The person upholds law and social rules to avoid the censure of the authorities and feelings of guilt about not doing one's duty. Maintaining social order is the goal.

78. You are asked to see a patient in the medical ward. He was clinically depressed and treated on Fluoxetine. His routine investigations showed sodium levels of 122 mol/L. Fluoxetine has now been stopped. Which of the following anti-depressants would be a suitable option in this situation?

A.Mirtazapine

B.Citalopram

C.Paroxetine

D.Amitriptyline

E.Moclobemide

Answer: E

This person has developed hyponatraemia (normal range is 136-145 m mol/L) following treatment with antidepressants such as Fluoxetine. For those who develop hyponatraemia with an SSRI, consider Noradrenergic drugs such as Nortriptyline and Lofepramine or an MAOI such as Moclobemide.

79. What type of EEG changes are routinely seen in Delirium Tremens?

a)Low voltage fast activity

b)Diffuse slowing of the background rhythm

c)Increased triphasic waves

d)Generalized periodic 1-2 Hz sharp waves

e)Focal delta activity.

Answer: A

Electroencephalography (EEG) can be useful in differentiating delirium from other conditions. The EEG shows a diffuse slowing of the background rhythm, in patients with delirium. In patients with delirium tremens, the EEG shows low voltage fast activity superimposed on slow waves.

80. A teacher rewarded Jimmy with star sticker for his good behaviour. It represents which type of learning?

A.Classical conditioning

B.Operant conditioning

C.Social learning

D.Aversive learning

E.Cognitive learning

Answer: B

Rewarding a child with star stickers for good behaviour is an example of positive reinforcement. It is said to take place when the behaviour is maintained or increased by producing consequences that the person finds rewarding. This is one of the principles used in operant conditioning.

81. A psychologist is involved in treating a patient with fear of heights. He requested the patient to imagine standing on top of the building without actual real exposure to them. The psychological principle employed here is;

A.Reciprocal inhibition

B.Aversion

C.Flooding

D.Implosion

E.Punishment

Answer: D

Flooding is an operant conditioning technique where actual exposure to feared stimulus takes place for a substantial amount of time so the accompanying anxiety response fades away while the stimulus is continuously present. This will lead to extinction of fear. When a similar technique is attempted with imagined not actual exposure then it is called as implosion.

82. Tom has history of severe alcohol dependence. He has been given Disulfiram to reduce the frequency of drinking alcohol. This is an example of;

A.Premack's principle

B.Avoidance learning

C.Aversive conditioning

D.Covert reinforcement

E.Covert sensitization

Answer: C

Aversive conditioning: This is an operant conditioning where punishment is used to reduce the frequency of target behaviour e.g. the use of Disulfiram (noxious stimuli) to reduce the frequency of drinking alcohol.

83. Which among the following EEG waves and their frequency is incorrectly matched?

A.Alpha waves- 8 to 13 Hz

B.Beta waves- more than 13 Hz

C.Theta waves- 7-11 Hz

D.Delta waves- < 4 Hz

E.Lambda- single waves

Answer: C

The frequency of Theta waves is 4 to 8 Hz. Mu waves occur over the motor cortex and their frequency is between 7-11 Hz

84. Which of the following can be used as an adjuvant to Risperidone to reduce Hyperprolactinaemia?

A.Sulpiride

B.Amisulpride

C.Trazadone

D.Naloxone

E.Aripiprazole

Answer: E

Hyperprolactinaemia and related symptoms are reported to improve fairly promptly following the addition of non-prolactin elevating drug such as Aripiprazole.

85. Which drug should be used with caution in patients with psoriasis?

A.Haloperidol

B.Lithium

C.Carbamazepine

D.Sodium valproate

E.Olanzapine

Answer: B

Lithium is involved in a variety of cutaneous reactions including psoriasis, which may present as exacerbation of preexisting psoriasis, induction of de novo psoriasis, pustular psoriasis, nail changes, and psoriatic arthropathy. The appearance of psoriatic lesions may occur at normal therapeutic serum lithium levels. The refractory period for the development of psoriatic lesions is variable and generally longer in induction and shorter in exacerbation of psoriasis. Lithium-induced psoriasis is often resistant to conventional treatment modalities, and some cases may require dose reduction or discontinuation of lithium treatment. (Ref: Jafferany M. Lithium and Psoriasis: What Primary Care and Family Physicians Should Know. Primary Care Companion to The Journal of Clinical Psychiatry. 2008;10(6):435-439)

86. Lucy is taking antipsychotic medications and complained of sexual dysfunction, menstrual disturbances and galactorrhoea. Which dopamine pathway when affected by antipsychotic drugs can induce such symptoms?

A.Mesocortical pathway

B.Mesolimbic pathway

C.Niagrostriatal pathway

D.Tuberoinfundibular pathway

E.All of the above

Answer: D

Lucy is exhibiting features of raised prolactin levels. Hyperprolactinaemia is associated with a number of adverse consequences such as sexual dysfunction, menstrual disturbances, breast growth, galactorrhoea and reductions in bone mineral density. A reduction in the dopaminergic input to the lactotroph cells results in a rapid increase in prolactin secretion. Such a reduction in dopamine can occur through the administration of antipsychotics, which acts on dopamine receptors, especially D2 in the tuberoinfundibular pathway of the brain.

87. Huntington's disease is a progressive neurodegenerative disorder, which is characterized by;

A.CCA > 35 expansion

B.CCA < 35 expansion

C.CTG > 35 expansion

D.CAG > 35 expansion

E.CAG< 35 expansion

Answer: D

Huntington's disease (HD) is a progressive neurodegenerative disorder with autosomal-dominant inheritance. The disease is caused by a CAG trinucleotide repeat expansion located in the first exon of the HD gene. The CAG repeat is highly polymorphic and varies from 6 to 37 repeats on chromosomes of unaffected individuals and from more than 30 to 180 repeats on chromosomes of HD patients. (Ref: Möncke-Buchner E, Reich S, Mücke M, et al. Counting CAG repeats in the Huntington's disease gene by restriction endonuclease *Eco*P15I cleavage. *Nucleic Acids Research*. 2002;30(16):e83)

88. Paul has been diagnosed with frontotemporal dementia. He has a strong family history. A mutation on which of the following chromosome predisposes a person to develop this condition?

A.5

B.7

C.17

D.19

E.22

Answer: C

Frontotemporal lobar degeneration (FTLD) refers to the 3 different syndrome of frontotemporal dementia (FTD), progressive non fluent aphasia and semantic dementia. Some patients with FTLD show tau protein based pathological changes. In familial cases, mutations have been identified in the microtubule-associated protein tau gene (MAPT) on chromosome 17q21.

89. A 26-year-old woman on psychotropic medications presents to the A&E department in a confused state. On examination, she has fever, raised BP, excessive sweating and bilateral clonus. The most likely diagnosis is;

A.Neuroleptic malignant syndrome

B.Serotonin syndrome

C.Antidepressant discontinuation syndrome

D.Hyponatraemia

E.Munchausen syndrome

Answer: B

Serotonin syndrome is a rare but potentially fatal syndrome occurring in the context of initiation or dose increase of a serotonergic agent, characterized by altered mental state (confusion), agitation, tremor, shivering, diarrhea, hyperreflexia, clonus, ataxia and hyperthermia.Autonomic disturbance - namely Hypertension, Tachycardia, Hyperthermia and Excessive sweating are also common.

90. Rosy has been diagnosed with generalized anxiety disorder. Which drug is licensed in the treatment of GAD?

A.Gabapentin

B.Topiramate

C.Lamotrigine

D.Pregabalin

E.Aripiprazole

Answer: D

Pregabalin is licensed in the treatment of GAD. Several large RCTs have demonstrated its efficacy and tolerability and comparable speed of onset of action to a benzodiazepine.

91. A 33-year-old woman is taking paroxetine for a few months. She developed pins & needles and shock like sensations in her legs. The diagnosis is;

A.Serotonin syndrome

B.Neuroleptic malignant syndrome

C.Antidepressant discontinuation syndrome

D.Akathisia

E.Dyskinesia

Answer: C

Antidepressant discontinuation symptoms include the following; Irritability, nausea, ataxia, diaphoresis, paresthesia and increased dreaming. The onset of symptoms is usually within 5 days of stopping treatment and it depends on the half-life of the antidepressant. The risk is increased in those prescribed short half-life drugs such as paroxetine and venlafaxine.

92. Phencyclidine acts on which receptor to produce symptoms of schizophrenia?

A.AMPA

B.Kainate

C.NMDA receptor

D.Dopamine receptor

E.Serotonin receptor

Answer: C

Phencyclidine blocks the activity of the NMDA receptor. It is a NMDA receptor antagonists that induces both positive and negative symptoms of Schizophrenia in healthy volunteers. This is possibly due to the modulation of the dopamine system and therefore exacerbates symptoms of patients with Schizophrenia.

93. John has depressive disorder. He is skeptical about taking antidepressants as he had sexual side effects with them. Which of the following has the lowest risk of sexual dysfunction?

A.Sertraline

B.Mirtazapine

C.Venlafaxine

D.Agomelatine

E.Trazadone

Answer: D

With Agomelatine, there is no clear effect on orgasm, erection or libido. The incidence of sexual dysfunction is similar to placebo.

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PAPER A RECALL- JUNE 2016 (Part 2)
by SPMM Administrator - Monday, 21 November 2016, 7:24 PM

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35. At your CAMHS inpatient unit, you observe Paolo, a four-year-old boy playing with his dinosaur toy. He treats it as though it is a living object. According to Piaget's model, which stage of cognitive development is he in?

a)Sensorimotor

b)Concrete Operational

c)Pre-operationial

d)Post-operational

e)Formal operational

Answer: C

This child is displaying animism, which is the belief that plants, inanimate objects and natural phenomena possess a living soul and have human feelings and intentions. The preoperational stage ranges from about ages 2 to 7 (Piaget, 1951, 1952). The child in this stage is pre (before) operations. This means the child cannot use logic or transform, combine or separate ideas. (Ref. Simply Psychology)

36. Pauline is a first-time mother to Stephanie, her 18-month-old daughter. She is worried that Stephanie isn't talking as well as the other children at the nursery. If an 18-month-old child has normal development, how many words should he or she know at this stage?

a)0- 10

b)11-25

c)25-50

d)50-85

e)85-125

Answer: C

In normal child development, an 18-month-old child's vocabulary will usually contain 20-50 words.

37. Which of the following is an example of an ideology that emphasizes the rights of groups over the individual?

A.Socialism

B.Democracy

C.Communism

D.Individualism

E.Collectivism

Answer: E

Group rights, also known as collective rights, are rights held by a group rather than by its individual members, the latter of which is termed individualism.

38. During Fresher's Week, Kaley, a 20-year-old law student, with no prior psychiatric history, presents to Accident and Emergency with delirium, labile blood pressure, dry mouth and dilated pupils. Which of the following could explain her presentation?

A.Anti-muscarinic psychosis

B.Neuroleptic malignant syndrome

C.Serotonin Syndrome

D.Paracetamol overdose

E.Cocaine intoxication

Answer: E

Cocaine is a stimulant made from an alkaloid contained in the leaves of the coca bush, Erythroxylon coca. Crack or rock cocaine is a ready-to-smoke, potent, relatively inexpensive, and highly addictive freebase forms of cocaine. The patient in question is presenting with typical symptoms of cocaine intoxication. (Ref. University of Southern California).

39. James comes to your memory clinic with a two-month history of short-term memory loss. He is concerned that he has Parkinson's disease as his late father had suffered from the disease. Which of the following genes is associated with Parkinson's disease?

a)Dysbindin

b)Neuregulin

c)Alpha synuclein

d)Presenilin-2

e)Amyloid Precursor Protein

Answer: C

Mutations in the following genes; alpha-synuclein (SNCA), parkin (PRKN), leucine-rich repeat kinase 2 (LRRK2), PTEN-induced putative kinase 1 (PINK1) are known to cause Parkinson's disease.

40. Toby is a 12-year-old boy in his first year of secondary school. His form tutor complains that he is extremely disruptive in school; jumping on desks and pushing in the cafeteria queue. You have reviewed him in your outpatient clinic and believe him to be suffering from ADHD. You arrange a meeting with his parents to discuss commencing medications. What is the mechanism of action of methylphenidate in treating ADHD?

A. Catecholamine re-uptake inhibition

B. Serotonin re-uptake inhibition

C. Dopamine agonist

D. Norepinephrine-Dopamine reuptake inhibitor

E. Sigma-1 antagonist

Answer: D

Methylphenidate is a phenethylamine stimulant that binds and blocks norepinephrine and dopamine transporters. The levels of these two hormones are thought to be low in the pre-frontal cortex of individuals with ADHD and, as a consequence, affect executive function (inhibiting behaviours, reasoning, organizing etc). Methylphenidate increases the firing rate of norepinephrine and dopamine, which ultimately enhances executive functioning.

41. Casper Williams is a 25-year-old postgraduate student studying business administration. He attends the student mental health service complaining of memory loss, which is affecting his ability to study. He worries that he will fail his MBA. On exploration of his social history, he admits to frequently using angel dust. Phencyclidine-induced amnesia is due to action on which of the following receptors?

A. Serotonin

B. Acetyl choline

C. Noradrenaline

D. Dopamine

E. Glutamate

Answer: E

Phecyclidine (PCP, angel dust) primarly acts as an antagonist at ionotropic glutamate receptors such as NMDA.

42. Tom has severe anxiety disorder and recently started on Pregabalin. His partner, who is a chemistry graduate, is keen about its receptor action. Pregabalin acts on which of the following receptors?

A. Glutamate

B. Serotonin

C. Acetyl choline

D. Noradrenaline

E. GABA

Answer: E

Pregabalin is a GABA analogue structurally (similar to gabapentin). Like gabapentin, pregabalin binds to the α2δ subunit of the voltage-dependent calcium channel in the central nervous system. This may subtly reduce release of certain neurotransmitters. It may as well influence GABA-ergic neurotransmission. It has anti-epileptic, analgesic (neuropathic pain) and anxiolytic effects. It is more potent than gabapentin hence has a higher therapeutic index and fewer dose-related side effects.

43. Hyder is a 35-year-old male on your assessment ward. He has a long-standing history of depression for which he takes citalopram. He complains of pain in his leg having slipped and fallen over on the ward the previous morning. He comes to you asking for painkillers. Which of the following increases GI bleeds with SSRIs?

A. Paracetamol

B. Diclofenac

C. Tramadol

D. Ibuprofen gel

E. Codeine

Answer: B

SSRIs inhibit the serotonin transporter, which is responsible for the uptake of serotonin into platelets. It is thought that SSRIs deplete platelet serotonin, leading to a reduced ability to form clots and a subsequent increase in the risk of bleeding. Non-steroidal anti-inflammatory drugs, such as ibuprofen and diclofenac, are another class of drug that causes gastrointestinal bleeding (Ref. BMJ 2005).

44. Mohammed is a 7-year-old boy. His mother told him that he could play outside once he had finished his homework. This is based on which of the following psychological mechanisms?

A. Reciprocal inhibition

B. Classical conditioning

C. Cognitive mapping

D. Operant conditioning

E. Premack's principle

Answer: E

This is Premack's principle, which states that high frequency behaviour can be used to reinforce low-frequency behaviour. This is also known as 'Grandma's rule' - 'eat the carrot and get the dessert'. This is positive reinforcement but instead of a material reinforcer, an already existing behaviour (eating dessert) is linked to a desired response (eating carrots).

45. Empathy skills are most likely to be delayed in which of the following?

A. A deaf child of deaf parents

B. A deaf child of hearing parents

C. A hearing child of deaf parents

D. A hearing child of hearing parents

E. A hearing child with one deaf parent

Answer: B

Deaf children with deaf parents have access to communication between signing adults. Deaf children brought up by hearing parents are at a disadvantage as there is less signing in this group.

46. Which of the following is seen in a normal child development?

A. Babbling with consonant sounds at 7 months

B. Social smile at 6 months

C. Toilet training at 5 years

D. Putting two words together at 4 years

E. Beginning to crawl at 15 months

Answer: A

Babbling is the production of articulate sounds that are not yet recognizable words. It begins shortly after birth. At around 7 months, babbling contains alternations of vowels and consonants e.g. baba, mama (Ref. NHS)

47. Which of the following is NOT a risk factor for delayed speech development?

a) Absence of a father

b) Rapid successive pregnancies

c) Small family size

d) Low levels of parental education

e) Smoking in pregnancy

Answer: C

The important family risk factors associated with delayed language development in children include large family size, smoking in pregnancy, rapid successive pregnancies, absence of father, low levels of parental education, divorce, marital disharmony & violence.

48. In the WAIS -R test battery for intelligence, which sub-test is resistant to brain damage/injury?

a) Digital span

b) Digit symbol

c) Similarities

d) Block design

e) Picture completion

Answer: E

Tests on WAIS are either 'hold' or 'no hold'. The so-called hold tests in WAIS are thought to reflect the use of old knowledge and are relatively resistant to the effects of brain damage and ageing. These include vocabulary, information, object assembly, and picture completion. No hold tests require speed of response, working memory, or the creation of new relations between unrelated items. These are more likely to show early decline with ageing and cognitive impairment. These include digit symbol, digit span, similarities, and block design. This is the reason why WAIS is not the best test to measure premorbid intelligence.

49. Which of the following has the strongest correlation with developmental language delay?

a) Maternal smoking during pregnancy

b) Maternal alcohol use during pregnancy

c) Low socio-economic status of mother

d) High fibre diet

e) Immigration

Answer: A

Smoking during pregnancy and large family size are the strongest risk factors for developmental language delay

50. Clive is a 27-year-old skateboard artist with paranoid schizophrenia. He was commenced on clozapine two months ago. He attends his GP complaining of severe chest pain. His GP takes recordings of his vitals and performs an ECG. What is likely to be seen on Clive's ECG?

A. ST depression

B. ST elevation

C. Prolonged QT

D. Tall peaked u waves

E. Pathological q waves

Answer: B

Clive is most likely suffering from myocarditis, an inflammation of the heart that can be caused by clozapine. ECG findings include: diffuse T wave inversions and saddle-shaped ST segment elevations.

51. According to the DSM-5, which of the following is true regarding changes to the diagnostic criteria of psychotic illnesses?

A. All subtypes of schizophrenia were removed

B. Simple schizophrenia is not a formal diagnosis

C. Schizo-typal disorder is now classified amongst schizophrenia

D.Scizhopherniform illness is no longer classified amongst schizophrenia

E.Catatonia is strongly associated with schizophrenia.

Answer: A

The DSMV is the 2013 update to the American Psychiatric Association's (APA) classification and diagnostic tool. Changes were made in the diagnostic criteria and coding of Schizophrenia spectrum and other psychotic disorders. This included the removal of all subtypes of schizophrenia (paranoid, disorganized, catatonic, undifferentiated, and residual).

52. According to the World Health Organization, which of the following constitutes the strongest risk factor for domestic violence?

A.Being female

B.Partner misuses substances

C.Social deprivation

D.Low social economic status

E.Low education

Answer: A

The key risk factor for being a victim of all forms of domestic and gender-based violence is being female (Ref. WHO)

53.A psychology student is keen to know about the ICD-10 duration criteria for the diagnosis of Mania. The episode should last at least;

A.1 day

B.2 days

C.1 week

D.2 weeks

E.4 weeks

Answer: C

Both ICD-10 and DSM-4 specify the manic episode should last at least 1 week and the disturbance is sufficiently severe enough to impair social and occupational functioning.

54.The Nerve Supply of Superior Oblique muscle is;

A.Oculomotor

B.Trochlear

C.Trigeminal

D.Facial

E.Abducent

Answer: B

The Superior oblique is the only extra ocular muscle innervated by the trochlear nerve.

55.Sophie had a review in the outpatients' clinic for her depressive disorder. The Psychiatrist is keen to start her on Agolematine. Which of the following baseline investigation should be done before starting her on this drug?

A.Complete blood count

B.Liver function test

C.Kidney function test

D.Thyroid function test

E.ECG

Answer: B

Liver toxicity is a side effect of Agomelatine. Liver function test should be performed in every patient before starting him or her on this drug. LFTs should be monitored at 3, 6, 12 and 24 weeks after starting treatment.

56.Which of the following drugs may exacerbate the symptoms of Schizophrenia due to NMDA receptor antagonistic activity?

A.Cannabis

B.Cocaine

C.Amphetamine

D.Ketamine

E.LSD

Answer: D

NMDA receptor antagonists like Ketamine and Phencyclidine have been shown to induce both positive and negative symptoms of Schizophrenia in healthy volunteers. This is possibly due to the modulation of the dopamine system and therefore exacerbates symptoms of patients with Schizophrenia.

57.Which of the following is the biggest risk factor for developing NMS?

A.Older age

B.Female gender

C.Starting on anticholinergic medications

D.Afrocarribean ethnicity

E.Sudden withdrawal of antiparkinsonian medication

Answer: E

The syndrome of NMS has been described in all age groups. Age is not a risk factor. Men outnumber women 2 fold. Sudden withdrawal of antiparkinsonian medication is one of the high risk factors for developing NMS.

58.Which of the following structures is involved in the pathophysiology of Wernicke's encephalopathy?

A.Hippocampus

B.Prefrontal cortex

C.Amygdala

D.Basal ganglia

E.Mamillary bodies

Answer: E

In Wernicke's encephalopathy, pathological lesions are seen in mammillary bodies, thalamic nuclei and the walls of the third ventricle.

59.A grandmother says to her grandchild "You must eat your greens before you go to play outside". Which psychological mechanism is used here?

A.Premack's principle

B.Reciprocal inhibition

C.Counterconditioning

D.Overt sensitization

E.Classical conditioning

Answer: A

Premack's principle is also called relative theory of reinforcement, often used with young children. It states that more probable behaviour (like play outside) will reinforce less probable behavior (like eating green leafy vegetables)

60.'**Sawtooth wave pattern'** in EEG is seen in which of the following phases of sleep?

A.Stage 1 n-REM sleep

B.Stage 2 n-REM sleep

C.Stage 3 n-REM sleep

D.Stage 4 n-REM sleep

E.REM sleep

Answer: E

The sawtooth wave is a distinctive pattern in electroencephalogram (EEG) that occurs only in REM sleep and that is recorded mainly in the central scalp area.

61. Which of the following antipsychotic is a highly selective dopamine antagonist?

A.Chlorpromazine

B.Haloperidol

C.Olanzapine

D.Clozapine

E.Sulpride

Answer: E

Amisulpride and Sulpride have highly selective dopamine antagonistic property.

62. Benzodiazepines enhance the effects of which of the following neurotransmitters?

A.Dopamine

B.Serotonin

C.GABA

D.Acetyl choline

E.Norepinephrine

Answer: C

It is widely believed that benzodiazepines enhance the effects of the neurotransmitter gamma-aminobutyric acid (GABA). They do so through action at benzodiazepine receptors.

63.Which class of drugs should be avoided in organic disinhibition?

A.Sodium valpraote

B.Olanzapine

C.Benzodiazepines

D.SSRIs

E.Trazadone

Answer: C

Benzodiazepines should be avoided in organic disinhibition. There is some evidence that SSRIs and Trazadone may ameliorate disinhibition in fronto-temporal dementia

64. Which of the following deletion syndromes is considered as the highest known genetic risk factor for schizophrenia other than having a monozygotic twin sibling with the illness?

A.5q 11.2

B.14 q 11.2

C.19 q 11.2

D.22 q 11.2

E.25 q 11.2

Answer: D

People with 22q11.2 deletion syndrome (velo-cardio-facial syndrome) have a 30-fold risk of developing schizophrenia. During adolescence and early adulthood, up to 30% of people with the syndrome develop schizophrenia, thus suggesting that the 22q11.2 deletion can be considered the highest known genetic risk factor for schizophrenia other than having a monozygotic twin sibling with the illness.

65. An ECG should be carried out on presentation in a patient with anorexia nervosa if BMI is less than;

A.16

B.19

C.20

D.22

E.25

Answer: A

In patients with anorexia, a screening assessment of risk should include a minimum of blood pressure, pulse, BMI, temperature and tests of proximal myopathy. An electrocardiogram (ECG) should be carried out if weight is low (BMI <16). Where a low BMI combines with abnormalities on ECG, rapid gradient of weight loss, abnormal electrolytes and muscle wasting, these can be considered 'rough and ready' predictors of higher physical risk. ECG is recommended routinely if the BMI is <16, if drugs affecting QT interval are prescribed or if there are electrolyte abnormalities

66. Which of the following conditions is associated with polycystic ovarian syndrome?

A.Anorexia nervosa

B.Bulimia nervosa

C.Body dysmorphic disorder

D.Binge eating disorder

E.Muscle dysmorphia

Answer: B

Absent follicular development is seen in 20% of women with bulimia and bulimia is associated with polycystic ovarian syndrome.

67. The strongest treatable risk factor for vascular dementia is;

A.Diabetes

B.Hypertension

C.Hypercholesterolemia

D.Atrial fibrillation

E.Obesity

Answer: B

The Syst-Eur and PROGRESS studies showed small but significant reductions in dementia from treating hypertension (Forette 2002; Tzourio 2003). Hypertension is now considered as strongest treatable risk factor for vascular dementia.

68.What is the most common psychiatric diagnosis among those presenting to emergency departments following self-harm?

A.Schizophrenia

B.Depression

C.Panic disorder

D.Adjustment disorder

E.Post traumatic stress disorder

Answer: D

The most common psychiatric diagnosis among those presenting to emergency departments following self-harm is Adjustment disorder.(Ref: Taggart et al; Gen Hosp psychiatry 2006: 28: 330-5)

69. The gene most strongly implicated in ADHD is

A.Dopamine 1 receptor gene

B.Dopamine 2 receptor gene

C.Dopamine 3 receptor gene

D.Dopamine 4 receptor gene

E.Dopamine 5 receptor gene

Answer: D

The gene most strongly implicated in ADHD is the D_4 dopamine receptor gene DRD4. The distribution of DRD4 mRNA in the brain suggests that it plays a role in cognitive and executive functions. These functions are implicated in the pathophysiology of ADHD (Patterson 1999)

70. CADASIL is associated with mutation in which of the following genes?

A.Parkin gene

B.Notch3

C.COMT

D.LRRK2

E.Dysbindin

Answer: B

An unusual form of subcortical small vessel disease occurs in the rare inherited condition of CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leuko-encephalopathy) in which a mutation in the Notch3 gene results in deposition of granular material in the walls of small arteries and arterioles, narrowing of vessel lumens and destruction of smooth muscle cells in their walls. (Ref: Vascular dementia: a pragmatic review; BJPsych Advances; Sept 2012;372-380)

71.The symptoms of adjustment disorder resolve within;

A.1 week of termination of the stressor

B.2 weeks of termination of the stressor

C.1 month of termination of the stressor

D.3 months of termination of the stressor

E.6 months of termination of the stressor

Answer: E

Adjustment disorder requires a stressor and that the symptoms should resolve within 6 months of termination of the stressor or its consequences.

72.Which one among the following is an environmental factor associated with ADHD?

A.Food and additive allergies

B.Low-protein

C.High-carbohydrate diets

D.Prematurity

E.Essential fatty acid deficiencies

Answer: D

The environmental risk factors associated with ADHD include Prenatal and/or perinatal risks (e.g. prematurity, low birth weight, exposure to toxins), Parental and family factors (e.g. child neglect, inconsistent parenting, high expressed emotion, family conflict, early institutional rearing) and Acquired neurobiological risks (e.g. head trauma) Ref: Asherson 2005. All others listed above are biochemical risk factors for ADHD.

73. Mutations in the progranulin gene are known to cause;

A.Creutzfeldt Jakob disease

B.CADASIL

C.Parkinson's disease

D.Lewy body dementia

E.Frontotemporal dementia

Answer: E

Mutations in the progranulin gene are known to cause frontotemporal lobar degeneration, leading to dementia (autosomal dominant type).

74.A child attains gender identity in which stage of Freud's psychosexual development?

A.Oral phase

B.Anal phase

C.Phallic phase

D.Latent phase

E.Genital phase

Answer: C

In Phallic phase between the ages of 3-7, the child learns to realize the differences between males and females and becomes aware of its own sexuality.

75. The preferential attachment of a baby to one adult female figure is called;

A.Secure attachment pattern

B.Proximity seeking

C.Monotropy

D.Universal attachment

E.Separation distress

Answer: C

Bowlby has been criticized for emphasizing Monotropy, the view that the child initially forms attachment to one caregiver, usually the mother. Even when several carers are available, infants appear to show a clear preference for the primary caregiver. Bowlby termed this tendency of the infant to form one special attachment 'Monotropy'.

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Paper A Recall - June 2016
by SPMM Administrator - Sunday, 20 November 2016, 8:30 PM

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PAPER A RECALL- JUNE 2016

1.Mrs Davis is a 55-year-old woman who has been taken to A&E by her daughter. She was found to be behaving increasingly oddly over the past few days, isolating herself, suspicious of the neighbours, preoccupied with ideas that she is being persecuted. Further investigation reveals that, following a recent Deep Vein thrombosis, the GP had abruptly stopped one of her usual prescribed medications. Which one of the following prescribed medications is most likely to be implicated in the above scenario?

A)Testosterone

B)Androgen

C)Estrogen

D)Progesterone

E)Thyroxine

Answer: C

Hormone Replacement Therapy is frequently given to post-menopausal women. Estrogen is a powerful psychoactive hormone, with widespread physiological effects. It is believed that its psychoactive mechanism is mediated by endorphins. There are reports indicating that sudden withdrawal of estrogens has precipitated mood disturbances and psychosis-like presentations.

2.Ms. Triggs has a diagnosis of Bipolar Affective Disorder Type II, with two previous hypomanic episodes and numerous depressive episodes. Her mood is currently low, and she is experiencing insomnia, increased appetite, and negative ruminations. You are considering starting an antidepressant. Which of these is least likely to precipitate a manic switch?

A)Fluoxetine

B)Sertraline

C)Bupropion

D)Amitryptiline

E)Venlafaxine

Answer: C

Tricyclics like Amitryptiline and Venlafaxine in particular carry a greater risk of precipitating a switch into mania. SSRIs have also been implicated; there is some evidence for bupropion also contributing to manic switch, but this evidence is less robust than for the other agents listed.

3.Jack Lyndon is a 43-year-old man with a long history of depression, who one month previously had been started on an antidepressant he's not previously tried. He has found some alleviation in his depressive feelings, his hypersomnia and lethargy. He is invited to a dinner party and part-way through the evening he begins to feel unwell, with facial flushing. The host offers to measure Mr. Lyndon's blood pressure with a portable machine, and it is found to be 208/106. Which of the following is most likely to be related to the hypertensive crisis?

A)Tranylcypromine

B)Fluoxetine

C)Mirtazapine

D)Sertraline

E)Agomelatine

Answer: A

When MAOIs such as Tranylcypromine are taken, the patient must be warned to avoid foods (such as mature cheese and red wine) containing tyramine (a monoamine precursor), as it can precipitate a hypertensive crisis, which is potentially life-threatening.

4.William and Diane Corndale are seeking help for infertility. It is 5 years since they have been trying to conceive, but have so far been unsuccessful. Which of the following is the most likely consequences of this stressor?

A)Negative impact on self-esteem

B)Decrease in frequency of sexual intercourse

C)Increase in number of disagreements

D)Feelings of guilt

E)All the above

Answer: E

Difficulty in conceiving may place significant stress on a couple, and there's the potential for a wide spectrum of psychological struggles to ensue.

5.Your colleague Bill asks you to call a GP practice on his behalf. He reads out loud the number for you, and you manage to remember it long enough to dial it. What kind of memory does this illustrate?

A)Semantic memory

B)Working memory

C)Implicit memory

D)Episodic memory

E)Declarative memory

Answer: B

Working memory is the ability to process and remember information at the same time, and usually hold small amounts of information (typically around 7 items). It is how we process information such as phone numbers, which we can remember sufficiently long to perform the task of dialling it.

6. Mrs Molloy has been struggling with feelings of depression and self-blame. She's been talking about her childhood experiences in psychotherapy and described the way her father would make her follow very strict rules. On one occasion, she returned home only a few minutes after her curfew, and she was beaten by him with a belt. The parenting style she is describing is known as

   A)Permissive

   B)Indulgent

   C)Neglectful

   D)Authoritarian

   E)Secure

Answer: D

Authoritarian parenting tends to be restrictive and rule-focussed. It is designed to teach the child to be submissive and fearful, to cope with a harsh world. Children generally tend to grow up with little sense of agency or personal power, with a tendency towards depression, self-doubt and fear.

7. Mr Cox is a 56-year-old man with chronic schizophrenia, who has been smoking 20 cigarettes a day since his teenage years. On a routine medical examination, it is noted that the BP in his right arm is 184/98, and on the left 155/80. The examination is repeated on several other occasions, and the difference in blood pressure between the arms is found to be consistent. Which artery is most likely to be affected?

   A)Subclavian artery

   B)Middle cerebral artery

   C)Posterior inferior cerebral artery

   D)Cerebellar artery

   E)Radial artery

Answer: A

The vignette suggests that the subclavian artery is affected, most likely by atherosclerosis. Discrepancy in left and right arm blood pressures is associated with higher risk of cardiovascular events.

8. Mrs Dellmore, who is 83 years old, has been referred to a clinic for older adults following concerns that she has been forgetting the names of common objects, and losing her way on returning from the local shops. The psychologist uses a number of rating measures. Which one is observer-rated?

   A)PHQ-9

   B)GAD-7

   C)NPI-C

   D)SADQ

   E)Meyers Briggs

Answer: C

The Neuropsychiatric Inventory- Clinician rating scale is observer-rated, focussing on neuropsychiatric symptoms in people with dementia. The other scales are self-rated (A and B for depression and anxiety, D for alcohol, and Meyers Briggs for personality types).

9.Brian Hughes is a war veteran who has been diagnosed with PTSD. He reports that his sleep is particularly disrupted and undergoes investigation at a sleep clinic. Which of the following findings is most consistent with sleep and PTSD?

A)Increased frequency of rapid eye movements in REM sleep

B)Longer duration of stage 2 NREM sleep

C)Longer total sleep time

D)Decreased sleep onset latency

E)Decreased stage 1 NREM sleep

Answer: A

The effect on sleep in people with PTSD is the focus of much research and there have been several findings; the most consistent is that in those with PTSD (compared with controls), there is an increase in stage 1 NREM sleep, decreased Slow Wave Sleep (SWS) and increased average number of rapid eye movements per minute in REM sleep

10.Mr. Welton's memory is being tested. He has been given a list of items to study. Once he has finished studying the items he is given another list to study, before being asked to recall as many items as he can from the first list, in any order. What memory test does this allude to?

A)Semantic Memory

B)Immediate Free Recall

C)Delayed Free Recall

D)Iconic Memory

E)Working Memory

Answer: C

This test includes a delay prior to the free recall of items on the first list; hence it is 'delayed' free recall.

11.Timmy has a pet mouse which he spends time with. As part of a psychological experiment, his father creates a loud noise whenever Timmy is playing with his mouse, and the noise frightens Timmy. This is done a number of times, so that whenever Timmy sees his pet mouse, he feels frightened. Which of these is now classed as the conditioned stimulus?

A)The mouse

B)The noise

C)The mouse + noise

D)The fear

E)The fear + noise

Answer: A

The mouse was previously a neutral stimulus; it has been associated with an unconditioned stimulus (the noise) that naturally triggers an automatic response in Timmy (fear). The mouse therefore becomes a conditioned stimulus, eliciting fear in Timmy.

12.You are about to go to A&E to assess a patient who is feeling suicidal. Which demographic description has got the strongest association with completed suicide?

A)Elderly male

B)Elderly female

C)Young female

D)Middle-aged male

E)Widow

Answer: D

Rates of suicide in men is nearly four times higher than that of women, with middle-aged men at highest risk, with a suicide rate in the UK of 26.5 per 100,000.

13. Mr Loughton has come to see you in clinic, following a referral from the Police. You are aware that your colleague will want to know what his future risk of sexual violence will be. Which of the following is the strongest predictive factor for future sexual violence?

A)History of alcohol misuse

B)History of depression

C)Having numerous siblings

D)History of past sexual violence

E)History of bipolar affective disorder

Answer: D

History of past sexual violence is the strongest predictor for future sexual violence (Scott 2004)

14. Mrs Swami is about to start taking Olanzapine for a newly diagnosed psychotic illness. Her nephew is reading up on Olanzapine's metabolism and modes of action. What property of a drug such as Olanzapine aids its movement into fat tissues?

A)It induces Cytochrome CYP450

B)It is lipophilic

C)It is metabolised in phase I oxidative metabolism

D)It is highly protein-bound

E)It is renally excreted

Answer: B

Lipophilic molecules travel well across the phospholipid bilayer of the cell membrane. They are nonpolarized and nonionized. Greater the lipophilicity, higher the movement of drugs into fat tissues.

15. Angela Stubbs has been participating in psychotherapy for the past seven months. She has the sense that the process has been helping her develop a more robust sense of self, a feeling of continuity, and of affiliation with others. What term best describes this process?

A)Identity formation

B)Identity foreclosure

C)Identity moratorium

D)Identity diffusion

E)Identity fusion

Answer: A

Identity formation or individuation is a process in which the sense of self becomes more distinct; this can be accompanied by an increased ability to engage with others, pursue life goals and gain greater satisfaction from life.

16. Sammy Taylor is coming to see you in clinic as an emergency book-in. He is 19 years old and has developed some voice-hearing experiences and bizarre thoughts regarding the MI5. Which of these might be a precipitating factor?

A)Having been adopted at the age of 7 months

B)Having a brother who has a diagnosis of schizophrenia

C)Having lost his adoptive father to a road traffic accident two weeks previously

D)Having had a traumatic birth

E)Having dropped out of school prematurely at the age of 15

Answer: C

The others may be contributing factors, but most are predisposing rather than precipitating. A precipitating factor has a strong temporal link to the onset of the episode of mental illness (e.g. it occurs just before).

17.Thomas Wilkins is a 56-year-old man who presented to his GP with some personality changes, including anxiety, depression, confusion and memory loss. His movements lacked coordination and he began to experience falls. There was a suspicion of vCJD and a head MRI was arranged. Which of the following signs gives an accurate diagnostic sign for vCJD (variant Creutzfeldt-Jakob disease)?

A)Hypothalamus hyperintensity

B)Pulvinar sign

C)Frontal lobe atrophy

D)Cerebellar hypointensity

E)Pineal gland sign

Answer: B

The Pulvinar sign (symmetrical hyperintensity in the pulvinar (posterior) nuclei of the thalamus) on brain MR images is a highly accurate sign for vCJD.

18.Andrew is a 10-day-old genetic male who appears not to have developed the characteristic of male genitalia as expected. Biochemical tests are performed. The cause is also linked to excessive virilisation in female adults. Which condition does this relate to?

A)17-beta-hydroxysteroid dehydrogenase deficiency

B)17-alpha-hydroxysteroid dehydrogenase deficiency

C)5-alpha-reductase deficiency

D)5-beta-reductase deficiency

E)3-beta-yydroxysteroid dehydrogenase

Answer: A

17-beta-hydroxysteroid dehydrogenase deficiency is an autosomal recessive form of male pseudohermaphroditism due to impaired testicular conversion of androstenedione to testosterone. It can produce impaired virilization of genetically male infants & children and excessive virilization of female adults. It affects male sexual development and their bodies do not produce enough of testosterone.

19.Michael Stills is 32 years old and has developed depression for the first time in in his life. You meet him and take a history. Which of the following is a precipitating factor for his depression?

A)Being married for the past 2 years

B)The loss of his baby son from cot death at the age of 4 months

C)Growing up in a large city

D)Working as a teaching assistant

E)Having a parent who has a diagnosis of chronic anxiety disorder

Answer: B

Bereavement is strongly associated with the onset of depression.

20. Mrs Mokolo has been reading up on the biochemistry of SSRIs, after you started her on Sertraline for an episode of depression. She reminds you that the serotonergic receptor with auto-regulatory function in synaptosomes (5-HT terminals) is:

A)5-AT1A

B)5-HT1B

C)5-HT2A

D)5-HT2B

E)5-HT1D

Answer: B

It is the 5-HT1B auto-receptors that are located in serotonin terminals. 5-HT1B auto-receptors regulate 5-HT reuptake, as well as modulating 5-HT release and synthesis.

21. Which of the following statements about dopamine D2 receptors is NOT true?

A)They affect adenylyl cyclase activity

B)They are excitatory

C)They act via the tuberoinfundibular pathway

D)They constitute both pre and post-synaptic receptors

E)They are G protein-coupled

Answer: B

D2 are inhibitory receptors that have both pre- and post-synaptic functions. They inhibit adenylyl cyclase activity and act via the tuberoinfundibular pathways that inhibit prolactin. Antipsychotics remove the inhibitory control of dopamine by blocking D2 receptors in tuberoinfundibular tract. This leads to hyperprolactinaemia, which presents as gynaecomastia in males and galactorrhoea in females. Long-standing prolactin increase may lead to osteoporosis.

22. Homosexuality is no longer considered as a mental disorder. Which edition of the *International Classification of Diseases* (ICD) saw the removal of homosexuality as a mental illness?

A.ICD 1

B.ICD 2

C.ICD 3

D.ICD 9

E.ICD 10

Answer: E

The World Health Organisation's ICD-9, published in 1977, listed homosexuality as a mental illness. It was later removed from the ICD-10, a decision that was endorsed by the 43rd World Health Assessment on 17th May 1990.

23. George Smith, a 74-year-old retired teacher, presents to your outpatient clinic with his wife, Mary. She has expressed her concern that George has been spending an increasing amount of time in the betting office. His medical history includes; hypertension, Parkinson's disease, depression and gastrooesophageal reflux disease. Of the following medications he is taking, which is the most likely cause of pathological gambling?

A.Zopiclone

B.Levodopa

C.Omeprazole

D.Metoprolol

E.Mirtazapine

Answer: B

Impulsive and compulsive behaviour is a possible side effect of some medications used to treat Parkinson's disease, including levodopa. Although rare, it can have a profound effect on the patient and their family. It is important to discuss this risk with patients when prescribing medication for Parkinson's disease. (data from Parkinson's UK)

24. Which of the following medications acts on NMDA receptors?

A.Memantine

B.Mirtazapine

C.Methodone

D.Tranylcypromine

E.Moclobemide

Answer: A

Antidementia medication, Memantine, works by blocking N-methyl-d-aspartate (NMDA) glutamate receptors. Unlike ketamine, which is a high affinity noncompetitive blocker, memantine is a non-competitive blocker with low affinity and binds only to actively open NMDA channels. Its receptor dissociation rate is relatively fast and so it does not accumulate and interfere with normal NMDA activity.

25. You are asked by the matron in A&E to review Sarah Moore, a 43-year-old mother of two, who separated from her husband, John, earlier this year. For the past few months, Sarah has been spreading lies about John to other parents, including mistruths that he is taking their two children out of school during school hours. This resulted in a meeting between John and the school welfare officer. Sarah appears happy at this conclusion and tells you that it proves she is the better parent. These false accounts appear to coincide with John's new relationship. What is the name given to this particular phenomenon?

A.Confabulation

B.Falsification

C.Pseudologia fantastica

D.Déjà vu

E.Depersonalisation

Answer: C

Pseudologia fantastica is the production of convincing false accounts often with apparent sincere conviction. There may be a grandiose or over-exaggerated flavour to the accounts produced. It is a feature of Munchausen's disease. (Ref. Oxford Handbook of Psychiatry). Their (incessant) lies are mostly aimed at attracting noise and inflating their importance (Delhi Psychiatry Journal 2007)

26. In a strange situation experiment, Bobby, a 2-year-old boy, is only comforted when his mother returns to the room and is not comforted by the stranger. What type of attachment is seen here?

A. Secure attachment

B. Anxious Avoidant

C. Anxious Resistant

D. Disorganized

E. Insecure

Answer: A

Secure attachment: Ainsworth constructed a strange situation experiment with 2 separation and 2 reunion episodes. The most common type of attachment behaviour in the Western world is secure, which constitutes 70% of infants. Here, the child will play independently when the mother is in the vicinity (secure base effect) and show distress when the mother leaves. He / she will seek contact on the return of the mother and be comforted quickly by her, and not the stranger.

27. In your CAMHS clinic, you observe Mollie, a five-year-old girl, as she plays with her doll. You notice that she is talking to her doll as though it is real. What is the name of this phenomenon?

a)Artificialism

b)Symbolic play

c)Egocentrism

d)Animism

e)Centration

Answer: D

Animism is the belief that plants, inanimate objects and natural phenomena possess a living soul and have human feelings and intentions. Piaget identified four stages of animism: a) Up to the ages of 4 or 5 years, the child believes that almost everything is alive and has a purpose. b) During the second stage (5-7 years) only objects that move have a purpose. c) In the next stage (7-9 years), only objects that move spontaneously are thought to be alive. d) In the last stage (9-12 years), the child understands that only plants and animals are alive. (Ref. Child, Adolescent and Family Development 2012)

28. Mandip is a 33-year-old Sikh male from India. He is an IT contractor and speaks English at work. He regularly goes to the pub after work for drinks with his colleagues. When he goes home he speaks Punjabi with his parents and wears traditional robes. What form of acculturation does Mandip display?

A.Marginalisation

B.Assimilation

C.Integration

D.Separation

E.Division

Answer: C

Acculturation refers to the process of cultural change that takes place when an individual or a group comes in to continuous contact with a culturally distinct group. Acculturation can result from immigration and can occur in either direction - hosts can get accultured; this was evident in certain places in times of Colonial rule. Four types are described according to the degree of retention and adoption of the two cultures at ethnocultural group level. Integration refers to both high retention of one's own cultural values and high adoption of the practices of the new culture.

29. You are discussing clozapine treatment with Luke, a 24-year-old male with treatment refractory paranoid schizophrenia. Which of the following lifestyle factors will produce an increased level of clozapine.

A.He plans to stop smoking by his next birthday in two months' time

B.He doesn't drink coffee or tea

C.He stopped taking fluvoxamine six years ago following a single episode of depression.

D.He drinks on average four pints of lager every week

E.He is taking propanolol for anxiety

Answer: A

Clozapine plasma concentrations can rise 1.5 times in the 2-4 weeks following smoking cessation (de Leon, 2004) and in some instances by 50-70% within 2-4 days. The effects of inhibitors (caffeine) are seen sooner than those of inducers (smoking), which require CYP1A2 synthesis (de Leon, 2004).

30. Which of the following is an aspect of pharmacokinetics?

A.Neurotransmitter synthesis

B.Neurotransmitter storage

C.Storage release

D.Reuptake

E.Absorption

Answer: E.

Pharmacokinetics is concerned with the time course and disposition of drugs in the body (what the body does to the drug). It involves the processes of Absorption, distribution, Metabolism and Elimination (ADME). Contrast with pharmacodynamics, which refers to the study of the mechanism of action of drugs (the effect of drugs on the body). Most psychotropics affect neurotransmitters of the brain. This effect can occur at various levels.

31. In his 1987 book, *Le Suicide*, what did sociologist Emile Durkheim conclude?

A.Suicide could be attributed to a lack of social capital

B.Suicide rates are higher in women than in men

C.Suicide was more common during times of war

D.Suicide was more common in civilians than in soldiers

E.The risk of suicide is fixed

Answer: A

Durkheim noticed that suicide rates were higher among Protestants (where integration levels were low) than Catholics and Jews (where integration is normal) arguing that stronger social control among Catholics resulted in lower suicide rates.

32. Shalinee, a 30-year-old florist, has a history of epilepsy for which she takes carbamazepine. She is planning to start a family with her husband, David and seeks your advice on medication during pregnancy. Which of the following pharmacokinetic changes occurs during pregnancy?

A.Reduced volume of distribution

B.Increased plasma protein binding

C.Reduced absorption from intramuscular site of injection

D.Increased renal elimination

E.Reduced rate of metabolism

Answer: D

Gastric emptying and small intestine motility are reduced in pregnancy due to elevation of progesterone, which tends to reduce absorption via the oral route. Total body water increases by up to 8 litres, creating a larger space within which hydrophilic drugs may distribute. Total plasma concentration of albumin-bound drugs decreases as a result of haemodilution and the subsequent fall in plasma albumin concentration and reduced plasma protein binding. Some enzymes of the hepatic cytochrome P-450 system are induced by oestrogen/ progesterone, resulting in a higher rate of metabolism (and hence elimination) of drugs. Renal blood flow is increased by 60-80% during pregnancy, and glomerular filtration rate rises by 50%, leading to enhanced elimination of drugs (Ref. Dawes 2001)

^

33. During a cognitive examination, Betty, an 85-year-old former shop assistant is asked to read a telephone number and then dial the number without looking at it. What type of memory is being tested here?

A.Episodic memory

B.Iconic memory

C.Haptic memory

D.Recognition memory

E.Semantic memory

Answer: B

Iconic memory is the memory of visual stimuli. It is part of the visual memory system that includes visual short-term memory and long-term memory. It is a sensory memory that is thought to last milliseconds before disappearing.

34. Betty is then shown a list of words. After a few moments she is shown another list of words and asked to identify words that appear in both lists.

A.Episodic memory

B.Iconic memory

C.Haptic memory

D.Recognition memory

E.Semantic memory

Answer: D

Recognition memory is the ability to recognize previously encountered events, objects or people.

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Economic Analysis Video
by SPMM Administrator - Sunday, 10 April 2016, 2:40 AM

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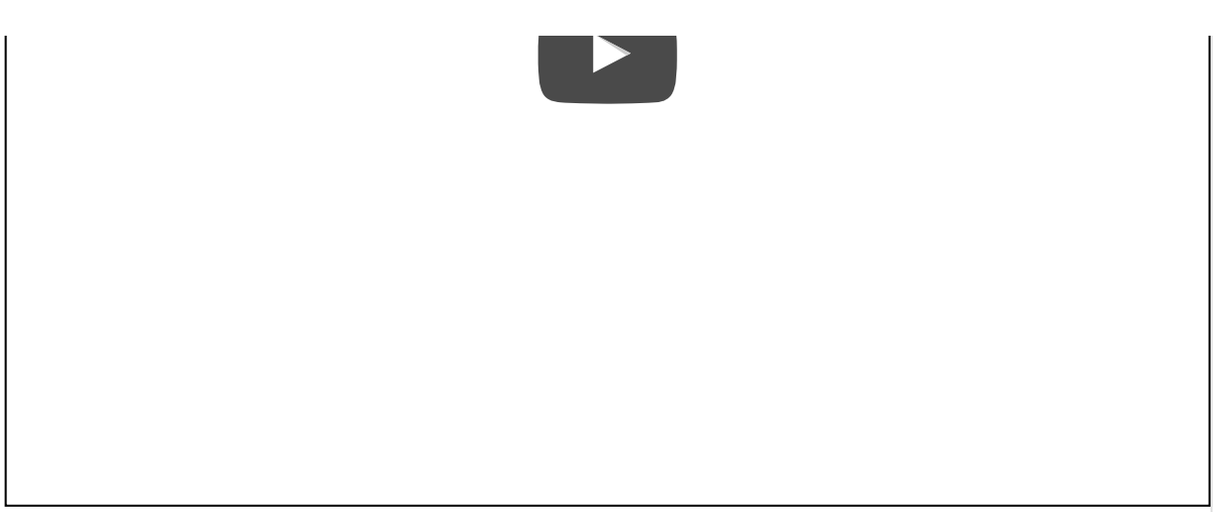
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Economic Analysis (part 2) Graphs





All the best for your exams. Always feel free to email us if you have any queries,

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[Modified: Sunday, 10 April 2016, 2:44 AM]
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Updated schizophrenia genetics table
by SPMM Administrator - Sunday, 5 July 2015, 3:35 PM

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The table below is adapted from Harrison (2015) Journal of Psychopharmacology 2015, Vol. 29(2) 85-96. This summarizes the most recent genetic findings in schizophrenia from GWAS and CNV studies.

Locus	Gene	Notes
Single Nucleotide Polymorphisms		
12p13.33	CACNA1C (L-type calcium channel)	Important for neuronal function; mutations cause Timothy syndrome and Brugada syndrome
12q24.11	D-amino acid oxidase	Enzyme that degrades d-serine (NMDA co-agonist)
1q42.2	DISC-1 (Disrupted in schizophrenia-1)	Seen in a Scottish family with 1:11 translocation
11q23.2	Dopamine D2 receptor	Target for antipsychotic action
2q33-34	Receptor tyrosine kinase erbB4	Neuregulin 1 receptor
5q33.2	AMPA receptor subunit 1	Affects synaptic plasticity
16p13	NMDA receptor subunit 2A	Influences channel conductance and synaptic localisation
7q21	Metabotropic glutamate receptor 3	Inhibitory autoreceptor
1p21	Micro RNA 137	Regulates transcription
8p12	Neuregulin 1	Growth factor
17p13	Serine racemase	Synthesizes d-serine from l-serine
18q21	Transcription factor 4	Deletion causes Pitt-Hopkins syndrome
2q32	Zinc finger 804A	Affects gene regulation esp. in cortical pyramidal neurons
Copy Number Variations		
2p16.3 deletion	Neurexin 1	Involved in synaptic structure
7q36.3 duplication	Vasoactive intestinal peptide receptor 2	Regulates synaptic transmission in the hippocampus, and development of neural progenitor cells in the dentate gyrus
hemi deletion of 22q11	Involves COMT coding genes	velocardiofacial syndrome

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Using the HiYield Course

by SPMM Administrator - Sunday, 10 May 2015, 5:49 AM

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[Modified: Sunday, 10 May 2015, 5:51 AM]

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